

Assigning Parking Sensors to Displays — User Guide

This guide explains how to choose which parking sensors each guidance display counts, using the **Display Map Editor**, and how to load the result into the controller.

1. How it works

Each parking space has a **sensor**. Sensors are connected to the controller on **sensor lines (ports)**. Each **display** shows the number of free spaces among the sensors assigned to it.

Item	Range	Notes
Displays	0 – 9	10 displays
Ports (sensor lines)	0 – E	15 lines
Sensors on each port	0 – F	16 sensors per line

All numbers are hexadecimal. After 9 comes A, B, C, D, E, F. So port A is the 11th line and sensor F is the 16th sensor on a line.

A sensor can count toward several displays. For example, a sensor on level 1 can count toward the "Level 1" display and also toward the "Whole car park" display at the entrance.

2. Before you start

You need:

- **A computer with Microsoft Edge, Google Chrome or a recent Firefox.**
- **The editor**, started in one of two ways:
 - Double-click **start.bat** in the editor folder. A small minimized window (the local server) opens, and the editor opens in Edge. Keep the small window open while you work; close it when you are done.
 - Or open the editor's web address, if your company publishes it online.
- **To send the settings to the controller:** a USB-to-RS485 adapter wired to the controller's configuration RS485 line. The adapter must switch transmit/receive direction automatically.

Opening **index.html** directly by double-click also works for editing and saving files, but **sending to the controller only works when the editor is started with start.bat** (or from the web address).

3. The editor screen

From top to bottom:

1. **Config File:** buttons **Open**, **Save**, **Save As...**, **New**, and the **append saveconf** checkbox. The name of the open file is shown at the top right. An **orange dot** next to it means there are unsaved changes.

- 2. **Select Display**: one button per display (**Display 0 ... Display 9**). The small number on each button is how many sensors that display currently counts.
- 3. **Sensor Assignments**: the grid for the selected display. Rows are **Port 0 ... Port E**, columns are sensors **0 ... F**. A **blue cell** means "this sensor counts toward this display". The **Hex** column on the right shows each row as a 4-digit hex value.
- 4. **All Displays Overview**: all 10 displays and 15 ports at a glance as hex values. Click a row to edit that display.
- 5. **Config Text Preview**: the exact text that will be saved to the file or sent to the controller.
- 6. **Send to Device (Serial / RS485)**: connection and sending to the controller.

4. Step by step: assign sensors to a display

Step 1 — Open the existing configuration

Click **Open** (or press **Ctrl+O**) and choose the configuration file, for example **RS13cfg.txt**. You can also drag the file from Windows Explorer onto the editor page.

A green message confirms how many assignments were loaded. A red message lists lines the editor could not understand; those lines are ignored.

To start from nothing, click **New**. All assignments are cleared.

Step 2 — Select the display

Click the display button, for example **Display 0**. The grid title changes to **Sensor Assignments — Display 0**.

Step 3 — Mark the sensors

Use any of these methods:

To do this	Do this
Add or remove one sensor	Click its cell
Add or remove several sensors quickly	Press the mouse button on a cell and drag across other cells
Add or remove a whole port (all 16 sensors)	Click the Port name at the left of the row
Add or remove one sensor number on all ports	Click the sensor number (0 ... F) in the header
Enter a row directly	Type a 4-digit hex value in the Hex box and press Enter (see section 5)

The number on the display button updates as you work.

Step 4 — Repeat for the other displays

Select the next display and mark its sensors. Useful shortcuts:

- **Copy display / Paste display:** copies all assignments of the current display to another display. Handy for a "total" display: copy the first level, then add the other levels.
- **Clear display:** removes all assignments of the current display (you are asked to confirm).

Step 5 — Check the overview

Look at **All Displays Overview**. Each cell shows one port's value for one display. 0000 (shown dimmed) means no sensors from that port are counted. Click any row to jump back and edit it.

Step 6 — Save the file

Click **Save** (or press **Ctrl+S**).

- In **Edge** and **Chrome** the first save asks for a file name and location. After that, **Save** writes to the same file directly.
- In **Firefox** each save downloads a copy of the file to your Downloads folder.

Use **Save As...** to save under a different name, for example to keep a backup copy before changes.

Always keep a saved copy of the final configuration for each site.

5. Worked example

A car park has two levels and one entrance display:

Display	Shows free spaces for	Sensors
Display 0	Level 1	Port 0, sensors 0–F (16 spaces) and Port 1, sensors 0–7 (8 spaces)
Display 1	Level 2	Port 2, sensors 0–F (16 spaces) and Port 3, sensors 0–B (12 spaces)
Display 2	Whole car park (entrance)	Everything from Display 0 and Display 1

How to enter it:

1. Click **New**.
2. Select **Display 0**. Click **Port 0** (whole row). On **Port 1**, drag across sensors **0** to **7**.
3. Select **Display 1**. Click **Port 2**. On **Port 3**, drag across sensors **0** to **B**.
4. Select **Display 0** and click **Copy display**. Select **Display 2** and click **Paste display**. Then click **Port 2**, and drag across sensors **0** to **B** on **Port 3**.
5. Tick **append saveconf** and click **Save**.

The overview now shows:

	P0	P1	P2	P3	P4 ...
Disp0	FFFF	00FF	0000	0000	0000
Disp1	0000	0000	FFFF	0FFF	0000
Disp2	FFFF	00FF	FFFF	0FFF	0000

Reading the hex values

Each port value is the sum of the values of its marked sensors:

Sensor	0	1	2	3	4	5	6	7
Value	0001	0002	0004	0008	0010	0020	0040	0080

Sensor	8	9	A	B	C	D	E	F
Value	0100	0200	0400	0800	1000	2000	4000	8000

Common values:

Hex	Sensors
0000	none
000F	0–3
00FF	0–7
0FFF	0–B
FFFF	all (0–F)
0005	0 and 2

You never have to calculate these yourself. Clicking cells fills in the hex value automatically.

6. Send the configuration to the controller

1. Connect the USB-RS485 adapter to the computer and to the controller's RS485 configuration line.
2. In the **Send to Device** section, check the settings. Normally leave them as they are:

Setting	Default	Meaning
Baud / Parity / Data / Stop	9600, E, 8, 1	Line settings of the controller
Line end	CR	Character sent at the end of each command
Char timeout	10 ms	Longest wait for the controller to echo each character
Line delay	100 ms	Extra wait after each command for the controller's reply

3. Click **Connect** and choose the adapter's COM port in the browser window. The status changes to **Connected 9600 8E1** in green.

4. Make sure **append saveconf** is ticked if the controller should keep the settings after a power cycle (see below).
5. Click **Send config**. The green bar shows progress. Sending takes about half a minute. **Stop** interrupts sending.
6. When finished, check the line under the progress bar, for example:

Done: 2277/2277 bytes, 151 lines in 31.2s | echo OK 2277, no echo 0

- **no echo 0**: every character was confirmed by the controller. The transfer is good.
- **no echo** greater than 0: some characters were not confirmed. Check the wiring and settings, then send again.

The black log window shows everything the controller answered.

7. Click **Disconnect** before unplugging the adapter.

What **saveconf** does

The last command in the file, **saveconf**, tells the controller to store the new settings permanently. Without it, the controller uses the new assignments only until it is restarted.

- When you open a file that contains **saveconf**, the **append saveconf** box is ticked automatically, and **saveconf** stays at the end of the saved file.
- When the opened file has no **saveconf**, the box is unticked. Tick it yourself if you want to add it.

7. Configuration file format (for reference)

The file is plain text with one command per line. It can be viewed in Notepad.

```
setmapdp 0 0 FFFF
setmapdp 0 1 FF
setmapdp 0 2 0
...
setmapdp 9 E 0
saveconf
```

- **setmapdp <display> <port> <sensors>**: all three numbers are hexadecimal.
- Commands may be written in upper or lower case.
- Lines starting with **#** are comments and are ignored.
- Displays and ports that are not listed have no sensors assigned.

8. Troubleshooting

Problem	What to do
"Web Serial is not available in this browser"	Use Edge, Chrome or a recent Firefox, and start the editor with <code>start.bat</code> , not by opening <code>index.html</code> directly.
The adapter is not listed after clicking Connect	Check the USB cable and that the adapter's driver is installed (it should appear under <i>Ports (COM & LPT)</i> in Device Manager). Close other programs that may be using the port.
"no echo" is high, or the log shows garbage	Check A/B wiring of the RS485 line and the line settings (9600, E, 8, 1). Check that the adapter switches direction automatically.
Red message "unrecognized ..." when opening a file	The file contains lines in an old or unknown format. Those lines are skipped; check the assignments and save again to write a clean file.
"display/port out of range"	A line refers to a port above <code>E</code> or a display above <code>9</code> . Numbers in the file are hex: port <code>10</code> means 16, not 10.
Changes are lost after the controller restarts	Send again with append saveconf ticked.
The editor window warns about leaving the page	You have unsaved changes. Click Save first.